

Evidential Multi-Agent Orchestration for Multimodal Urban Flood Risk Assessment: A Subjective Logic Framework with Provenance-Aware Explanations

Ali Hamza^{a,*}, Ghulam Mujtaba^b

^aNational University of Sciences and Technology (NUST), Islamabad, Pakistan

^bNational University of Computer and Emerging Sciences (FAST-NUCES), Islamabad, Pakistan

ARTICLE INFO

Keywords:

Multimodal Machine Learning
Decision-Level Fusion
Subjective Logic
Multi-Agent Systems
Urban Flood Assessment

ABSTRACT

Urban flooding is among the most economically destructive categories of natural hazard, with the February–March 2022 south-east Queensland event ranking as one of Australia’s costliest flood episodes of recent decades. Effective operational decision-support requires fusing heterogeneous streams — satellite radar, multi-spectral optical imagery, reanalysis precipitation, terrain and land-cover layers, gauge and air-quality stations, and unstructured hydrological bulletins — that arrive at irregular cadences with frequent inter-source disagreement and mutual sensor outages. The downstream decision demands not only a class prediction but also a calibrated, auditable account of which sensors contributed, which were absent, and how strongly the system believes each candidate outcome.

1. Introduction

Urban flooding is among the most economically destructive categories of natural hazard, with the February–March 2022 south-east Queensland event ranking as one of Australia’s costliest flood episodes of recent decades. Effective operational decision-support requires fusing heterogeneous streams — satellite radar, multi-spectral optical imagery, reanalysis precipitation, terrain and land-cover layers, gauge and air-quality stations, and unstructured hydrological bulletins — that arrive at irregular cadences with frequent inter-source disagreement and mutual sensor outages. The downstream decision (issuing an inundation warning, prioritizing evacuation, allocating sandbag resources) demands not only a class prediction but also a calibrated, auditable account of which sensors contributed, which were absent, and how strongly the system believes each candidate outcome.

Three architectural families dominate current literature. LLM-orchestrated agentic pipelines delegate sub-tasks to specialist agents and arbitrate conflicts through prompt-chained reasoning [Jiang et al., 2026]. Remote-sensing-grounded multi-agent systems wrap tool-use and retrieval around geospatial foundation models [Talemi et al., 2026, Redaelli et al., 2026]. A third tradition couples dynamic knowledge graphs with environmental inference nodes to mediate between sensors and decision outputs [Hofmeister et al., 2024]. Parallel to this agentic wave sits an evidential-operator literature — multi-source Subjective Logic fusion [van der Heijden et al., 2018], partial-observable projection, credibility-weighted trust models, and evidential deep-learning heads — that is mathematically rigorous but not yet interfaced end-to-end with multi-agent environmental pipelines.

A close reading of these families exposes four structural limitations that block operational deployment. First, LLM-mediated arbitration produces uncalibrated probabilities: LLM-arbiters attain moderate ranking quality but collapse minority-class recall, generating a confident-but-wrong majority-class skew. Second, missing modalities are silently recovered or hallucinated, with no formal monotonicity guarantee on epistemic uncertainty under sensor dropout. Third, per-agent credibility is not propagated — a historically unreliable agent is weighted identically to a historically reliable one. Fourth, no source-provenance trail exists: post-hoc feature attribution explains feature importance but not which sensor or document caused a contribution, so an analyst cannot audit which data source was stale, missing, or uncalibrated. These four limitations motivate a single framework that combines specialist-agent decomposition, formal evidential reasoning, monotonic missing-modality projection, and source-level provenance.

We propose AEGIS¹ — an open-source framework: Agentic Evidential Geographic Intelligence for Sustainability, a specialist multi-agent framework in which each of five evidential agents — Climate, Satellite, Land-Cover, Air-Quality / Sensor, and Document Intelligence — emits a formal Subjective Logic opinion over a shared four-state Dirichlet frame. A deterministic coordinator inspects inter-agent agreement via a Jensen-Shannon signal. When agents agree, it routes through Consensus & Compromise Fusion; when they disagree, it routes through Weighted Belief Fusion weighted by per-agent credibility γ . Missing modalities automatically trigger a partial-observable projection that guarantees the epistemic uncertainty u grows monotonically as modalities drop, rather than subtracting false confidence. The cell-level fused opinion is concatenated with raw multimodal features and passed to a hybrid LightGBM evidential

*Corresponding author

Email address: ahamza.msse25mcs@student.nust.edu.pk (A. Hamza)

¹<https://github.com/ali-Hamza817/AEGIS>

head that retains the formal uncertainty signal as an input feature while preserving a competitively accurate prediction surface. Every emitted and fused opinion is logged to a DuckDB-backed provenance ledger with manifest IDs and source references, powering an interactive Leaflet dashboard that surfaces per-cell uncertainty, per-agent credibility, and per-source contribution shares.

The remainder of this paper is organized as follows. Section 2 reviews related work across agentic environmental AI, subjective-logic foundations, and conflict-resolution techniques. Section 3 develops the mathematical foundations of AEGIS. Section 4 details the architecture. Section 5 describes the Brisbane 2022 study, datasets, baselines, and metrics. Section 6 reports hypotheses H1–H4 with a mechanistic ablation isolating the contribution of the evidential machinery. Section 7 concludes with limitations, broader applicability to other environmental risk domains, and future work.

2. Related Work

The earliest formal treatment of disagreement between distributed agents was articulated by [Nielsen and Parsons \[2006\]](#), who showed that Bayesian networks held by separate agents can be fused through formal argumentation schemes, introducing an explicit syntax for managing inter-agent inconsistency. [Wang and Singh \[2007\]](#) then proposed the canonical Brier-score trust model that underpins nearly every subsequent reputation system in MAS, demonstrating that continuous track-record scoring outperforms binary endorsement under uncertainty. Together these two contributions fixed the foundational vocabulary still in use today: agents maintain beliefs, disagreements must be reasoned about formally, and trust must be tracked quantitatively.

The limitations of purely probabilistic MAS reasoning — inability to express honest “I don’t know,” no formal operator for missing observations, and no algebraically closed multi-source fusion — were addressed by the Subjective Logic operator algebra. [Kaplan et al. \[2015\]](#) introduced the partial-observable projection that mathematically guarantees epistemic uncertainty rises when evidence is absent, and [van der Heijden et al. \[2018\]](#) derived the Weighted Belief Fusion and Consensus & Compromise Fusion operators that compose an arbitrary number of opinions without losing algebraic closure. These contributions supplied the missing operator algebra: a framework in which trust-weighted multi-source fusion is provably defined and missing signals propagate as quantified uncertainty rather than false confidence.

A second parallel wave applied evidential reasoning at the discriminative learning layer. [Li et al. \[2024\]](#) introduced adaptive uncertainty-based learning for text-based person retrieval; [Liu et al. \[2025\]](#) proposed MERIT for multi-view evidential liver-fibrosis staging; and [Chen et al. \[2025\]](#) re-examined essential versus nonessential settings of evidential deep learning. In parallel, [Sun et al. \[2024\]](#) introduced T2MAC, which propagates per-agent credibility through selective communication engagement, and [Chikoti \[2025\]](#)

consolidated the conflict-resolution taxonomy within MAS. These works demonstrated that evidential reasoning could be deployed at both the symbolic-fusion layer and the discriminative learning layer, but the two traditions remained disconnected from physical environmental inference.

The first attempts to apply this machinery to physical environmental inference emerged through dynamic knowledge graphs. [Hofmeister et al. \[2023\]](#) demonstrated cross-domain flood-risk assessment for smart cities using a dynamic semantic substrate mediating between data sources and inference nodes. [Hofmeister et al. \[2024\]](#) extended this into an explicit semantic-agent framework for automated flood assessment, supplying the architectural intuition that a mediating ontology was required. The agent layer remained, however, in early form and did not yet integrate computable trust or monotonic missing-modality handling.

The agentic environmental-AI wave accelerated between 2025 and 2026. Foundational surveys catalogued the architecture space [[Sayyad et al., 2025](#), [Talemi et al., 2026](#)], and domain-specific deployments followed rapidly across fire and flood applications: [Bairam Zadeh et al. \[2025\]](#) proposed a conceptual high-level MAS for wildfire management; [Lee et al. \[2025\]](#) built multi-agent geospatial copilots for remote-sensing workflows; [Shabbir et al. \[2025\]](#) benchmarked tool-augmented agents through ThinkGeo; [Sandeep et al. \[2026\]](#) framed context-aware MAS for wildfire insights; [Palanca Cantonjos and Biswas \[2026\]](#) deployed AgroAskAI for smallholder-farmer queries; [Jiang et al. \[2026\]](#) introduced Flood-LLM, fusing multi-source urban data for property-level flood-risk assessment; and [Redaelli et al. \[2026\]](#) operationalised the SaferPlaces agentic AI for global flood-risk intelligence. Every one of these systems, however, relies on prompt-chained arbitration: an LLM receives agent summaries and emits a decision. None of them delivers calibrated uncertainty, monotonic missing-modality handling, per-agent credibility, or source-level provenance.

The present paper introduces AEGIS to address exactly these four unresolved structural limitations. By interfacing the agentic architectural intuitions [[Hofmeister et al., 2024](#), [Talemi et al., 2026](#)] with the formal Subjective Logic operator algebra [[Kaplan et al., 2015](#), [van der Heijden et al., 2018](#)], the canonical trust model [[Wang and Singh, 2007](#)], and the credibility-aware communication work of [Sun et al. \[2024\]](#) and [Chikoti \[2025\]](#), AEGIS fuses five specialist agents over a shared Dirichlet frame, propagates calibrated epistemic uncertainty under arbitrary sensor dropout, and exposes a per-cell provenance manifest through a DuckDB-backed dashboard. We show on the Brisbane February–March 2022 case study that the resulting framework matches monolithic accuracy (F1-Macro 0.7190 vs 0.7106) while uniquely providing calibrated uncertainty, missing-modality monotonicity, and per-agent reputational weighting.

3. Mathematical Foundations

3.1. Opinion algebra and Dirichlet bijection

We model the environmental state space over the frame of discernment $\mathbb{X} = \{\theta_{\text{dry}}, \theta_{\text{saturated}}, \theta_{\text{surface}}, \theta_{\text{inundation}}\}$, a four-state classification distinguishing progressively wetter hydrological conditions. A Subjective Logic opinion over $\mathbb{X}$ is the tuple $\omega = (\mathbf{b}, u, \mathbf{a})$, where $\mathbf{b} \in \mathbb{R}^4$ is the belief mass vector over the four states, $u \in \mathbb{R}$ is the epistemic uncertainty mass, and $\mathbf{a} \in \mathbb{R}^4$ is the prior base-rate vector. The opinion satisfies the closure constraint $\sum_{k=1}^4 b_k + u = 1$ with $b_k \geq 0$, $u \geq 0$, and $\sum_{k=1}^4 a_k = 1$. Given Dirichlet evidence parameters $\alpha = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ with total evidence weight $C = \sum_{k=1}^4 \alpha_k$ and non-informative prior constant W , the bijective mapping onto opinion space is

$$b_k = \frac{\alpha_k - C \cdot a_k}{C + W}, \quad u = \frac{W}{W + C} \quad (1)$$

and the projected expected probability over each state is

$$P(x_k) = b_k + a_k \cdot u \quad (2)$$

which reduces to a standard probability vector when $u = 0$ and to a diffuse vacuous distribution as $u \rightarrow 1$.

3.2. Partial-observable projection

When a sensor or modality yields no observation, the agent formally projects its evidence onto the unobserved dimensions rather than imputing synthetic values. Following the projection rule of Kaplan et al. [2015], the projected Dirichlet parameters are $\alpha_k^{\text{proj}} = C \cdot a_k$. This forces $b_k \rightarrow 0$ for all k and $u \rightarrow 1.0$, yielding the monotonicity property that the AEGIS framework exploits throughout:

Theorem. *When an observing agent loses one or more modalities, the projected epistemic uncertainty u is a strictly non-decreasing function of the count of missing modalities [Kaplan et al., 2015, van der Heijden et al., 2018].*

Operationally, this guarantee means missing data produces auditable quantification rather than silent hallucination — every dropout event is detectable through a monotonic u increase recorded in the provenance ledger, and is therefore distinguishable from confident-but-wrong prediction patterns.

3.3. Multi-source fusion: WBF and CCF

The coordinator composes a sequence of $N \geq 2$ agent opinions over the same frame through one of two compounding operators. The first is Weighted Belief Fusion, the credibility-weighted extension of the canonical two-source rule, applicable for N -ary aggregation when per-agent reputations are non-identity. The second is Consensus & Compromise Fusion, an averaging operator that is provably associative and commutative over the opinion lattice for any $N \geq 2$ [van der Heijden et al., 2018]. Operator selection is driven by inter-agent agreement, as detailed in Section

4. Both operators close the operator algebra over the tuple $\omega = (\mathbf{b}, u, \mathbf{a})$ and are selected such that the partial-observable projection invariant is preserved under composition.

3.4. Specialist-credibility update

Each specialist agent carries a Brier-score reputation $\gamma_i \in [\gamma_{\min}, 1.0]$ updated against the realised outcome online. For an agent emitting projected probabilities $\hat{p}_{i,k}$ against the one-hot realised state y_k , the score is

$$\text{BS}_i = \frac{1}{K} \sum_{k=1}^4 (\hat{p}_{i,k} - y_k)^2 \quad (3)$$

and the reputation is updated as

$$\gamma_i^{(t+1)} = \text{clip} \left(\gamma_i^{(t)} + \eta \cdot (1 - 2 \cdot \text{BS}_i), \gamma_{\min}, 1.0 \right) \quad (4)$$

with learning rate η and lower bound $\gamma_{\min}$. The clipping step enforces a non-zero agent vote inside WBF weighting even when an agent has scored poorly over many rounds — an empirically important property so an agent cannot be silently zeroed out and remove itself from disagreement-resolution channels. This dynamic reputation tracking is what differentiates the AEGIS framework from monolithic late-fusion alternatives; an unreliable agent's effective contribution to WBF decays over successive rounds without manual re-tuning [Wang and Singh, 2007].

3.5. Hybrid-head composite feature space

The evidential prediction head consumes a 27-dimensional composite feature vector assembled through a concatenation

$$\mathbf{F} = [\mathbf{b} \mid u \mid \mathbf{a} \mid \text{raw features}] \quad (5)$$

The first twelve dimensions encode the four-state opinion tensor carried from the SL coordinator; the remaining fifteen dimensions encode the raw multi-modal feature channels used for fitting (climate, SAR polarisations, NDWI, NDVI, terrain slope, PM2.5, embedded document features). Critically, the u mass is preserved as a 13th explicit feature rather than collapsed into a scalar probability. The LightGBM gradient-boosting model consumes the full 27-dimensional vector in a single-shot fit. The composite space allows AEGIS to simultaneously inherit the calibrated uncertainty of the opinion layer, achieve accuracy competitive with raw-feature monolithic alternatives, and propagate the missing-modality invariant end-to-end. In Section 6, we empirically decompose whether the SL opinion tensor and the credibility tracker carry their share of the marginal accuracy gain.

References

- R. Bairam Zadeh, A. Elmi, and V. Moghaddam. A conceptual high-level multiagent system for wildfire management. *IEEE Transactions on Geoscience and Remote Sensing*, 2025. doi: 10.1109/tgrs.2025.3559062.

- M. Chen, J. Gao, and C. Xu. Revisiting essential and nonessential settings of evidential deep learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025. doi: 10.1109/tpami.2025.3583410.
- S. Chikoti. Conflict resolution techniques in multi-agent systems: technical review. *European Mathematical & Statistical Journal*, 9:76–92, 2025. doi: 10.59573/emsj.9.2025.76.
- Felix Hofmeister, George Brownbridge, Mark Hillman, and Raoul Schleidt. Cross-domain flood risk assessment for smart cities using dynamic knowledge graphs. *Sustainable Cities and Society*, 99:105113, 2023. doi: 10.1016/j.scs.2023.105113.
- Felix Hofmeister, Jing Bai, George Brownbridge, and Raoul Schleidt. Semantic agent framework for automated flood assessment using dynamic knowledge graphs. *Data-Centric Engineering*, 5:e11, 2024. doi: 10.1017/dce.2024.11.
- J. Jiang, Y. Wang, and M. Manfredini. Flood-llm: an ai-driven framework for property-level flood risk assessment using multi-source urban data. *Sustainability*, 18:2957, 2026. doi: 10.3390/su18062957.
- Lance Kaplan, Murat Şensoy, Supratik Chakraborty, and Gary Cirincione. Partial observable update for subjective logic and its application for trust estimation. *Information Fusion*, 25:1–12, 2015. doi: 10.1016/j.inffus.2015.01.005.
- C. Lee, V. Paramanayakam, A. Karatzas, C. Karakizi, N. Dimitriou, A. Tzotskou, I. Gkotsis, I. Karpouzas, C. Karterouli, D. Karatzas, and P. Maragos. Multi-agent geospatial copilots for remote sensing workflows. In *Proceedings of IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025. doi: 10.1109/igarss55030.2025.11243915.
- Shuang Li, Chen He, Xiang Xu, and Qi Tian. Adaptive uncertainty-based learning for text-based person retrieval. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 3450–3458, 2024. doi: 10.1609/aaai.v38i4.28101.
- Y. Liu, Z. Gao, N. Shi, X. Wang, Y. Zheng, S. Zhang, and D. Shen. Merit: multi-view evidential learning for reliable and interpretable liver fibrosis staging. *Medical Image Analysis*, 99:103507, 2025. doi: 10.1016/j.media.2025.103507.
- Soren H. Nielsen and Simon Parsons. An application of formal argumentation: fusing bayes nets in mas. In *Proceedings of AAMAS*, 2006. URL <https://vbn.aau.dk/da/publications/78420bd0-9c2c-11db-8ed6-000ea68e967b>.
- N. A. Palanca Cantonjos and A. Biswas. Agroaskai: a multi-agentic ai framework for supporting smallholder farmers’ enquiries globally. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, page 41458, 2026. doi: 10.1609/aaai.v40i47.41458.
- T. Redaelli, P. Mazzoli, V. Luzzi, G. Boni, C. De Michele, A. Mosso, M. T. Brunetti, L. Piciullo, M. Rossi, A. Cesari, A. Arganani, M. Bottazzi, M. Ferneti, and J. Cevallos. Saferplaces agentic ai: democratising global flood risk intelligence for disaster risk reduction and management. *EGUsphere*, pages EGU26–6458, 2026. doi: 10.5194/egusphere-egu26-6458.
- A. Sandeep, S. Jayarathna, and S. Sandaruwan. Context-aware multi-agent architecture for wildfire insights. *Sensors*, 26:1070, 2026. doi: 10.3390/s26031070.
- N. Sayyad, H. Dave, M. Kathane, S. Almelkar, V. Jagtap, R. Lakra, and A. Jadhav. Agentic ai systems: a review of architectures, autonomy, and ethical implications. *Journal of Artificial Intelligence Systems*, 2025. doi: 10.64252/xgfrhd85.
- A. Shabbir, M. A. Munir, A. Dudhane, S. Khan, H. Cholakal, F. S. Khan, and G. S. Xia. Thinkgeo: evaluating tool-augmented agents for remote sensing tasks. *arXiv preprint arXiv:2505.23752*, 2025.
- Chen Sun, Zeyi Zang, Jiaying Li, Jian Wang, and Siu-Ming Yiu. T2mac: Targeted and trusted multi-agent communication through selective engagement and evidence-driven integration. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 15120–15128, 2024. doi: 10.1609/aaai.v38i13.29438.
- N. A. Talemi, J. Boone, and F. Afghah. Agentic ai in remote sensing: foundations, taxonomy, and emerging systems. *arXiv preprint arXiv:2601.01891*, 2026.
- Rens W. van der Heijden, Hendrik Kopp, and Frank Kargl. Multi-source fusion operations in subjective logic. In *Proceedings of 21st International Conference on Information Fusion (IEEE ICIF)*, pages 1–8, 2018. doi: 10.23919/icif.2018.8455615.
- Yonghong Wang and Munindar P. Singh. Formal trust model for multiagent systems. In *Proceedings of the 20th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 1584–1589, 2007. URL <https://ijcai.org/papers07/Papers/IJCAI07-250.pdf>.